

Especificação — QR Promo + E-commerce + Wizard de Campanha

Data: 2026-07-04

Status: Proposta de design (não implementado)

Branch de referência: Smart-Signage-Studio-Vx5

Objetivo

Definir a arquitetura e os contratos para:

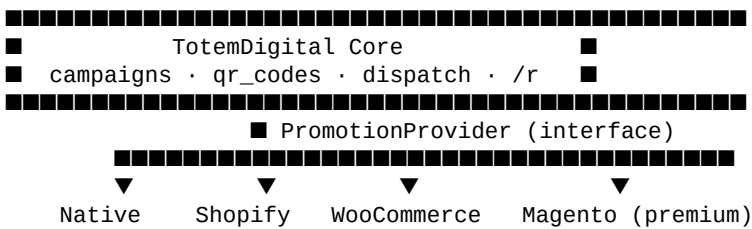
- Exibir **QR code com countdown** no totem (Player-AD), ligado a campanha/promo.
- Integrar lojas **Shopify** e **WooCommerce** (conectores padrão).
- Reservar **Magento** para tier **premium**.
- Wizard UI **“Usar cupom em campanha”** — do cupom da loja até o dispatch no totem.

O totem **nunca** processa pagamento. Checkout ocorre no celular do cliente após scan.

Decisões de produto

Decisão	Escolha
Conectores padrão	Shopify (OAuth) + WooCommerce (REST API keys)
Conector premium	Magento 2 (Integration token), gated por flag
Core (todos os planos)	QR + countdown + redirect <code>/r/{code}</code> + analytics
Render no player	WebView HTML (mediaType: qr_promo) — MVP
Escopo de integração	Por anunciante (subscriber_id), não global
Credenciais do app Shopify	Plataforma (system_settings), como SMTP

Camadas do produto



Camada	O quê	Cliente
Core	QR + countdown + redirect + scan_count	Qualquer anunciante
Connect Standard	Shopify ou WooCommerce	PME com loja existente

Camada	O quê	Cliente
Connect Premium	Magento 2	Enterprise, catálogo grande, B2B

Comparação rápida — Shopify vs WooCommerce vs Magento

Critério	WooCommerce	Shopify	Magento
API	REST WP + plugins	REST + GraphQL maduros	REST complexo, multi-store
Velocidade de integração	Média	Alta	Baixa (premium)
Autenticação	Consumer Key + Secret	OAuth 2.0	Integration token
Cupom / promo	/wc/v3/coupons	Discount codes API	Cart/catalog rules
Webhooks	Sim (WP)	Sim (nativo)	Sim
Melhor para	Cliente BR com WP/Woo	Integração rápida e estável	Enterprise

Ordem de implementação: Shopify → WooCommerce → Magento (premium).

Onde vive cada configuração

Escopo	Onde	Exemplo
Plataforma (admin)	system_settings	Client ID/Secret app Shopify TotemDigital
Anunciante	commerce_integrations	Token da loja loja-x.myshopify.com
Campanha / QR	qr_codes + commerce_promo_links	Cupom VERA030 → campanha #15

Diferente do **Financeiro** (SMTP/WhatsApp globais), e-commerce é **multi-tenant por subscriber**.

Schema proposto

Arquivos alvo (quando implementar):

- database/smartchannel-db-v2-refactored-part6-tables-other.sql
- database/smartchannel-db-v2-refactored-part7-foreign-keys.sql
- database/seeds-default-settings.sql

commerce_integrations

```
CREATE TABLE IF NOT EXISTS commerce_integrations (
  integration_id SERIAL PRIMARY KEY,
  subscriber_id INTEGER NOT NULL,

  provider TEXT NOT NULL, -- 'shopify' | 'woocommerce' | 'magento'
```

```

store_name TEXT,
store_url TEXT NOT NULL,
external_store_id TEXT,

status TEXT NOT NULL DEFAULT 'pending',
-- pending | connected | error | disconnected | revoked

access_token_encrypted TEXT,
refresh_token_encrypted TEXT,
api_key_encrypted TEXT,
api_secret_encrypted TEXT,
webhook_secret TEXT,

token_expires_at TIMESTAMP,
scopes TEXT[] DEFAULT '{}',

sync_enabled BOOLEAN DEFAULT true,
last_sync_at TIMESTAMP,
last_sync_status TEXT,
last_sync_error TEXT,

metadata JSONB DEFAULT '{} '::jsonb,
is_active BOOLEAN DEFAULT true,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

CONSTRAINT chk_commerce_provider
    CHECK (provider IN ('shopify', 'woocommerce', 'magento')),
CONSTRAINT chk_commerce_status
    CHECK (status IN ('pending', 'connected', 'error', 'disconnected', 'revoked')),
CONSTRAINT uq_commerce_subscriber_store
    UNIQUE (subscriber_id, provider, store_url)
);

```

commerce_promo_links

```

CREATE TABLE IF NOT EXISTS commerce_promo_links (
    link_id SERIAL PRIMARY KEY,
    integration_id INTEGER NOT NULL,
    campaign_id INTEGER NOT NULL,
    qr_id INTEGER,

    external_promo_type TEXT NOT NULL,
    external_promo_id TEXT NOT NULL,
    external_promo_code TEXT,
    title TEXT,
    expires_at TIMESTAMP,
    landing_url TEXT,

    sync_state TEXT DEFAULT 'active',
    metadata JSONB DEFAULT '{} '::jsonb,

    last_synced_at TIMESTAMP,
    is_active BOOLEAN DEFAULT true,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT uq_commerce_promo_external
        UNIQUE (integration_id, external_promo_type, external_promo_id)
);

```

commerce_sync_logs

```

CREATE TABLE IF NOT EXISTS commerce_sync_logs (
    log_id SERIAL PRIMARY KEY,
    integration_id INTEGER NOT NULL,
    action TEXT NOT NULL,
    status TEXT NOT NULL,
    details JSONB DEFAULT '{} '::jsonb,

```

```
        created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    );
```

Extensão em qr_codes (via metadata, sem ALTER obrigatório)

```
{
  "commerce": {
    "integration_id": 7,
    "provider": "shopify",
    "external_promo_id": "gid://shopify/DiscountCode/123",
    "coupon_code": "VERA030",
    "landing_path": "/discount/VERA030"
  },
  "utm": {
    "source": "totem",
    "medium": "qr",
    "campaign": "verao-2026"
  }
}
```

Evolução futura: adicionar qr_type = 'promo' ao CHECK de qr_codes (hoje: url, text, wifi, etc.).

Seeds — system_settings (plataforma)

Chave	Uso
commerce.shopify.client_id	App público TotemDigital na Shopify
commerce.shopify.client_secret	Secret (mascarado na UI)
commerce.shopify.scopes	read_discounts, read_products, read_price_rules
commerce.shopify.redirect_uri	https://{host}/api/commerce/shopify/callback
commerce.webhook_base_url	Base pública para webhooks
commerce.magento_premium_enabled	false por padrão
commerce.sync_interval_minutes	15
commerce.default_redirect_template	UTM padrão

Serviço espelhando financialIntegrationConfigService.ts: commerceIntegrationConfigService.ts
— resolve settings globais, cache ~30s, mascaramento de secrets.

Diagrama ER

```
erDiagram
    subscribers ||--o{ commerce_integrations : conecta_loja
    subscribers ||--o{ campaigns : anunciante
    commerce_integrations ||--o{ commerce_promo_links : sync_cupons
    campaigns ||--o{ commerce_promo_links : usa_promo
    campaigns ||--o{ qr_codes : QR
    commerce_promo_links ||--o{ qr_codes : gera_atualiza
    campaigns ||--o{ campaign_playlists : NM
    campaign_playlists ||--o{ playlists : ""
    playlists ||--o{ playlist_items : ordem
    playlist_items ||--o{ medias : midia
    campaigns ||--o{ campaign_totems : onde_exibe
    campaign_totems ||--o{ totems : ""
```

```

commerce_integrations ||--o{ commerce_sync_logs : auditoria

subscribers {
    int subscriber_id PK
    text name
}
commerce_integrations {
    int integration_id PK
    int subscriber_id FK
    text provider
    text store_url
    text status
}
commerce_promo_links {
    int link_id PK
    int integration_id FK
    int campaign_id FK
    int qr_id FK
    text external_promo_code
    timestamp expires_at
}
campaigns {
    int campaign_id PK
    int subscriber_id FK
    text title
    text status
}
qr_codes {
    int qr_id PK
    int campaign_id FK
    text code UK
    text redirect_url
    timestamp expires_at
    jsonb metadata
}
totems {
    int totem_id PK
}

```

Regra: o player lê apenas o **dispatch**; nunca chama API da loja diretamente.

OAuth — Shopify

```

sequenceDiagram
    participant U as Usuario
    participant FE as Frontend
    participant BE as Backend
    participant SH as Shopify OAuth

    U->>FE: Integracoes - Conectar Shopify
    FE->>BE: POST /api/commerce/shopify/connect
    BE->>BE: commerce_integrations status=pending
    BE->>BE: state=JWT(subscriberId, integrationId, nonce)
    BE-->>FE: authorizeUrl
    FE->>SH: Redirect OAuth
    U->>SH: Aprova app
    SH->>BE: GET /api/commerce/shopify/callback
    BE->>SH: POST /admin/oauth/access_token
    BE->>BE: token criptografado, status=connected
    BE->>SH: Registra webhooks
    BE-->>FE: Redirect /integrations?connected=shopify

```

Scopes mínimos (MVP): read_discounts, read_products, read_price_rules

Webhooks:

Tópico	Ação
discounts/update	Atualiza commerce_promo_links + qr_codes.expires_at
discounts/delete	sync_state = revoked, desativa QR
app/uninstalled	status = revoked, limpa tokens

WooCommerce — API keys (sem OAuth simples)

Fluxo wizard em 3 passos:

- URL da loja: `https://minhaloja.com.br`
- Consumer Key + Consumer Secret (Woo → Avançado → REST API, permissão **Leitura**)
- Testar: `GET /wp-json/wc/v3/coupons?per_page=5`
- `POST /api/commerce/woocommerce/connect`
`{ subscriberId, storeUrl, consumerKey, consumerSecret }`
- `POST /api/commerce/integrations/:id/register-webhooks` (opcional)
→ coupon.updated, coupon.deleted

Magento — tier premium

- Gated por `commerce.magento_premium_enabled` + plano do subscriber.
- `POST /api/commerce/magento/connect { storeUrl, accessToken }`
- UI com badge **Premium** se flag desligada.

UI — Settings e Integrações

Admin — nova aba E-commerce em Configurações

Permissão: `admin / platform_admin`.

- Client ID / Client Secret app Shopify (mascarado)
- Redirect URI (readonly, copiar)
- Scopes
- URL base webhooks
- Flag Magento premium
- Intervalo sync (minutos)

Anunciante — `/integrations` ou `/subscribers/:id/integrations`

- Cards: Shopify | WooCommerce | Magento (Premium)
- Lista lojas conectadas + sync manual + desconectar

- Lista cupons → botão **Usar em campanha**

API — rotas novas

Método	Rota	Quem
GET	/api/commerce/integrations	Anunciante (scope subscriber)
POST	/api/commerce/shopify/connect	Inicia OAuth
GET	/api/commerce/shopify/callback	Callback público
POST	/api/commerce/woocommerce/connect	CK/CS + teste
POST	/api/commerce/magento/connect	Premium
POST	/api/commerce/integrations/:id/test	Teste conexão
POST	/api/commerce/integrations/:id/sync	Sync manual
DELETE	/api/commerce/integrations/:id	Desconectar
GET	/api/commerce/integrations/:id/promos	Lista cupons
POST	/api/commerce/promo-links/wizard	Wizard passo 3
POST	/api/commerce/promo-links/:id/attach-playlist	Wizard passo 4
POST	/api/commerce/promo-links/:id/finalize	Wizard passo 5
POST	/api/commerce/webhooks/shopify/:integrationId	Webhook
POST	/api/commerce/webhooks/woocommerce/:integrationId	Webhook
GET	/r/:code	Redirect público (scan)

GET integration (sem secrets):

```
{
  "integrationId": 7,
  "provider": "shopify",
  "storeUrl": "loja-x.myshopify.com",
  "status": "connected",
  "lastSyncAt": "2026-07-04T12:00:00Z",
  "promoCountActive": 8,
  "credentialsMasked": { "hasAccessToken": true }
}
```

Wizard — “Usar cupom em campanha”

Entrada: **Integrações** → **cupom** → **Usar em campanha** ou **Campanha** → **Importar da loja**.

Passo 0 — Pré-requisitos

```
GET /api/commerce/integrations?subscriberId=42&status=connected
GET /api/campaigns?subscriberId=42&status=active,draft,approved
```

Passo 1 — Cupom

```
GET /api/commerce/integrations/7/promos?activeOnly=true&page=1&limit=20
```

```
{
  "success": true,
  "data": {
    "integrationId": 7,
    "provider": "shopify",
    "promos": [{
      "externalPromoType": "discount_code",
      "externalPromoId": "gid://shopify/DiscountCode/123456",
      "code": "VERA030",
      "title": "Verão -30%",
      "discountSummary": "30% off",
      "expiresAt": "2026-07-15T23:59:59Z",
      "usageLimit": 500,
      "usageCount": 127,
      "landingPath": "/discount/VERA030",
      "isActive": true
    }]
  }
}
```

Passo 2 — Campanha

Existente: GET /api/campaigns/15 — validar subscriberId.

Nova campanha rápida:

```
POST /api/campaigns

{
  "subscriberId": 42,
  "title": "Verão -30% Totem Shopping",
  "campaignType": "scheduled",
  "status": "draft",
  "startDate": "2026-07-04T00:00:00Z",
  "endDate": "2026-07-15T23:59:59Z",
  "priority": 5,
  "metadata": { "source": "commerce_wizard", "commerceProvider": "shopify" }
}
```

Passo 3 — QR + countdown

```
POST /api/commerce/promo-links/wizard
```

```
{
  "integrationId": 7,
  "campaignId": 15,
  "externalPromoType": "discount_code",
  "externalPromoId": "gid://shopify/DiscountCode/123456",
  "couponCode": "VERA030",
  "qr": {
    "code": "PROMO-VERA0-2026",
    "title": "Verão -30% — escaneie!",
    "subtitle": "Válido só hoje no shopping",
    "expiresAt": "2026-07-15T23:59:59Z",
    "maxScans": 500,
    "trackingEnabled": true,
  }
}
```

```
    "countdown": {
      "enabled": true,
      "position": "above",
      "format": "HH:mm:ss",
      "showWhenExpired": "message",
      "expiredMessage": "Promo encerrada"
    }
  },
  "utm": { "source": "totem", "medium": "qr", "campaign": "verao-2026" }
}
```

Response:

```
{
  "success": true,
  "data": {
    "linkId": 901,
    "qrId": 42,
    "code": "PROMO-VERAO-2026",
    "scanUrl": "https://totem.seudominio.com/r/PROMO-VERAO-2026",
    "renderHtmlUrl": "https://cdn.seudominio.com/qr/render/PROMO-VERAO-2026.html",
    "expiresAt": "2026-07-15T23:59:59Z",
    "serverNow": "2026-07-04T12:00:00Z"
  }
}
```

Backend (transação): upsert commerce_promo_links → insert/update qr_codes → gerar HTML/PNG.

Passo 4 — Playlist

POST /api/commerce/promo-links/901/attach-playlist

```
{
  "mode": "full_screen_item",
  "durationSeconds": 30,
  "playlistId": 88,
  "createPlaylistIfMissing": true,
  "playlistName": "Promo Verão QR",
  "insertPosition": "end"
}
```

Cria medias sintética (mediaType: qr_promo) + playlist_items.

Passo 5 — Totens + ativar

POST /api/commerce/promo-links/901/finalize

```
{
  "totemIds": [7, 12],
  "activateCampaign": true,
  "campaignStatus": "active"
}
```

Upsert campaign_totems, opcionalmente campaigns.status = active, invalidar cache dispatch.

Validações do wizard

Regra	HTTP
integration.subscriber_id === campaign.subscriber_id	403
Cupom expirado na loja	409 PROMO_EXPIRED

Regra	HTTP
Código QR duplicado	409 ou sufixo automático
Campanha sem contrato (se política exigir)	422 + aviso UI
Totem fora do escopo do publisher	403

Contrato dispatch — item `qr_promo`

Estende `DispatchMediaItem` existente (`backend/src/types/dispatcherTotem.types.ts`).

```
{
  "mediaId": 90001,
  "order": 4,
  "duration": 30,
  "url": "https://cdn.seudominio.com/qr/render/PROMO-VERAO-2026.html",
  "mediaType": "qr_promo",
  "cacheBucket": "propagandas",
  "mediaName": "QR Promo VERA030",
  "metadata": {
    "mimeType": "text/html",
    "qr": {
      "qrId": 42,
      "code": "PROMO-VERAO-2026",
      "campaignId": 15,
      "title": "Verão -30% — escaneie!",
      "subtitle": "Válido só hoje no shopping",
      "expiresAt": "2026-07-15T23:59:59Z",
      "serverNow": "2026-07-04T12:00:00Z",
      "maxScans": 500,
      "scanCount": 0,
      "scanUrl": "https://totem.seudominio.com/r/PROMO-VERAO-2026",
      "qrImageUrl": "https://cdn.seudominio.com/qr/PROMO-VERAO-2026.png",
      "countdown": {
        "enabled": true,
        "position": "above",
        "format": "HH:mm:ss",
        "showWhenExpired": "message",
        "expiredMessage": "Promo encerrada"
      },
    },
    "commerce": {
      "provider": "shopify",
      "integrationId": 7,
      "externalPromoId": "gid://shopify/DiscountCode/123456",
      "couponCode": "VERA030",
      "storeDomain": "loja-x.myshopify.com",
      "landingPath": "/discount/VERA030"
    }
  }
}
```

Campo	Função		
<code>expiresAt + serverNow</code>	Countdown sincronizado (calibra drift local)		
<code>scanUrl</code>	URL no QR — sempre domínio TotemDigital		
<code>duration</code>	Tempo na tela (ex.: 30s); ≠ validade da promo		
<code>showWhenExpired</code>	<code>hide \</code>	<code>message \</code>	<code>fallback_media_id</code>

Player-AD (MVP): tratar qr_promo como HTML/WebView (similar a html/widget em HtmlWebViewPlayback.kt).

Alternativa fase 2: overlay fixo via plan.metadata.overlays[].

Fluxo de scan

```
sequenceDiagram
    participant P as Player
    participant U as Cliente
    participant TD as TotemDigital /r/code
    participant L as Loja

    P->>P: QR + countdown
    U->>TD: GET /r/PROMO-VERAO-2026
    TD->>TD: expires_at, max_scans, is_active
    TD->>TD: scan_count++, event_log
    TD->>L: 302 redirect + UTM
    L->>U: Checkout / desconto
```

Redirect Shopify:

```
https://loja-x.myshopify.com/discount/VERA030
?utm_source=totem&utm_medium=qr&utm_campaign=verao-2026
&totem_id=7&qr_code=PROMO-VERAO-2026
```

Redirect WooCommerce:

```
https://minhaloja.com.br/?coupon-code=VERA030
?utm_source=totem&utm_medium=qr&utm_id=7
```

Worker de sync

```
commerceSyncWorker (intervalo: commerce.sync_interval_minutes)
FOR EACH integration WHERE status=connected AND sync_enabled
    syncPromos(provider)
        → upsert commerce_promo_links
        → atualiza qr_codes (expires_at, redirect_url, is_active)
        → commerce_sync_logs
```

Webhooks = invalidação imediata; job = fallback.

Estado atual no repositório (baseline)

Área	Status
Schema qr_codes	Existe (part6-tables-other.sql)
Backend qrcodeService.ts	Parcial; trechos legados
Frontend /qr-codes	Básico (nome + URL)
Player-AD QR	Não exibe — só vídeo/imagem/HTML
commerce_*	Não existe
Dispatch qr_promo	Não existe

Roadmap de implementação

#	Entrega
1	Schema commerce_* + seeds platform + commerceIntegrationConfigService
2	POST /api/commerce/promo-links/wizard + GET /r/:code
3	Wizard UI (passos 3→5) + página Integrações
4	OAuth Shopify + listagem cupons
5	WooCommerce connect + cupons
6	attach-playlist + dispatch qr_promo
7	Player-AD WebView qr_promo + countdown
8	Sync worker + webhooks
9	Magento premium

Referências no código

Área	Path
Dispatch types	backend/src/types/dispatcherTotem.types.ts
QR schema	database/smartchannel-db-v2-refactored-part-6-tables-other.sql
QR service	backend/src/services/qrcodeService.ts
Settings Financeiro (padrão secrets)	backend/src/services/financialIntegrationConfigService.ts
Settings UI	frontend/src/pages/Settings/Settings.tsx
QR UI (básica)	frontend/src/pages/QRCodes/QRCodes.tsx
Player HTML	Player-AD/.../HtmlWebViewPlayback.kt

Histórico

Data	Nota
2026-07-04	Documento inicial — design acordado em sessão de produto (QR + e-commerce)